MASTER ENGINEERING DESIGN DOSSIER

140.40 kWp Grid-Connected Rooftop Solar PV Installation

Al Noor Packaging Factory — Dubai, UAE

Prepared For: Al Noor Packaging LLC
Location: Al Quoz Industrial Area, Dubai, United Arab Emirates
Grid Connection Base: DEWA 3-Phase, 400 V, 50 Hz Utility Infrastructure
Engineering Framework: Shams Dubai DRRG Regulations Version 2.0 Compliance
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1. Phase 1 — Spatial & Structural Roof Suitability

A rigorous spatial analysis was performed on the reinforced-concrete roof deck of the primary manufacturing building to establish structural margins and physical boundary limits before design configuration.

Formula 1.1 — Gross Plan Window Envelope:

*A_gross = Length * Width = 62.0 m * 38.0 m = 2,356.0 m²*

Formula 1.2 — Net Usable Installation Area:

To ensure full safety access compliance, a mandatory 1.5-meter outer boundary setback corridor was subtracted from all perimeter faces. Additionally, unalterable mechanical water-tank housings (10m x 6m Eastern zone) and HVAC infrastructure groups (8m x 5m North-West zone) were completely removed from layout calculations:
A_net = A_gross - Setbacks - Mechanical Obstructions = 1,965.0 m²

Formula 1.3 — Structural Equipment Load Threshold:

The factory roof profile features a maximum pre-certified civil dead load capacity limit of 25.0 kg/m² for additional permanent mechanical equipment. The cumulative weight distribution of our optimized plant layout tracks safely beneath this ceiling:
Distributed System Mass = Modules (22.0 kg) + Racking + Ballast Block Matrix = 11.45 kg/m² < 25.0 kg/m² (SAFE)

Formula 1.4 — Dynamic Wind Design Pressure (q_z):

Per ASCE 7-22 codes tracking Dubai localized meteorological criteria, the extreme wind velocity envelope is calculated based on a 3-second gust design profile of 47 m/s, yielding:
*q_z = 0.613 * V² * K_z * K_zt * K_d = 1,197 N/m²*

2. Phase 2 — Electrical Sizing & Component Configuration

The facility's historical billing records present a gross cumulative energy import requirement of 1,098,000 kWh per annum. This represents a mean baseline demand profile of 3,008.2 kWh per day.

Formula 2.1 — Active Industrial Daytime Demand Baseline:

The manufacturing floor runs a robust operational shift from Monday to Saturday, between 08:00 and 18:00, compiling exactly 3,120 high-load daytime production hours annually. This maps out a steady instantaneous load floor:
P_daytime = Annual Generation Target / Operating Hours = 1,098,000 kWh / 3,120 h = 351.92 kW

Formula 2.2 — Idealized Offsetting Target vs. Infrastructure Constraints:

While a symmetrical 100% net bill offset requires a massive theoretical array capacity of 683.68 kWp under a localized regional 5.5 Peak Sun Hours (PSH) environment, the plant capacity is limited by facility connection

walls. The physical grid connection incomer contains a 630A main circuit breaker ceiling that imposes a maximum continuous solar power injection wall of 350 kW AC.

By deploying a single high-efficiency Huawei SUN2000-110KTL-M2 inverter unit (110 kW nominal capacity), the plant output profile aligns perfectly beneath the grid threshold. Furthermore, because the plant's maximum hourly generation (~110 kW) stays significantly below the active daytime base load (351.92 kW), 100% of the generated green energy is instantly consumed on-site. This satisfies the utility's mandatory zero-export / export limitation requirement and completely eliminates the risk of wasteful power dumping via active curtailment devices.

3. Phase 3 — Thermal String Capacity Boundaries

To shield the inverter's power electronic channels from overvoltage destruction while ensuring sufficient voltage to stay online during ambient climate extremes, thermal string length calculations were hand-calculated using the parameters of the selected LONGi 585W hardware asset.

Formula 3.1 — Maximum Winter Series String Length Safety Ceiling (N_max):

During worst-case cold winter mornings, regional ambient air drops to 5°C. As temperature decreases, the solar panel open-circuit voltage spikes inversely via its temperature coefficient (γ_{voc} = -0.23%/°C):
$V_{oc(Coldest)} = V_{oc(STC)} * [1 + (T_{min} - 25°C) * \gamma_{voc}] = 52.36V * [1 + (5 - 25) * (-0.0023)] = 54.36\ V$
Applying the rigid mathematical Floor function against the Huawei inverter's maximum allowed DC input envelope (1,100 V DC) locks in the string safety limit:
$N_{max} = Floor(V_{max_inverter} / V_{oc(Coldest)}) = Floor(1100\ V / 54.36\ V) = Floor(20.23) = 20\ Panels\ Max$

Formula 3.2 — Minimum Summer Series String Length Floor (N_min):

During peak summer heat waves, external air temperatures reach 50°C. Dark silicon absorption surfaces experience extreme thermal gains, driving internal cell heat up to 75°C (50°C ambient + 25°C radiant offset). This heat degrades module operating voltage:
$V_{mpp(Hottest)} = V_{mpp(STC)} * [1 + (T_{cell} - 25°C) * \gamma_{voc}] = 44.21V * [1 + (75 - 25) * (-0.0023)] = 39.12\ V$
To keep the strings generating full rated power, the combined rows must clear the Huawei inverter's minimum full-load MPPT tracking voltage requirement (540 V) even during peak summer heat waves:
$N_{min} = Ceil(V_{mpp_min_full_load} / V_{mpp(Hottest)}) = Ceil(540\ V / 39.12\ V) = Ceil(13.80) = 14\ Panels\ Min$

4. Phase 4 — Inter-Row Spacing & Aerodynamic Wind Analysis

To optimize rooftop space utilization and minimize wind sail vulnerabilities, the system is designed in a landscape (horizontal) orientation on single-row L1 click-fit frames. Turning the panels sideways shifts the physical dimension running up the incline slope to the module's width parameter (1.134 m) rather than its length (2.278 m).

Formula 4.1 — Inter-Row Shading Pitch Boundaries:

To guarantee zero inter-row shading at solar noon during the winter solstice (December 21st, solar altitude angle α = 41.5° where tan(41.5°) = 0.8847), row spacing is governed by trigonometric parameters:

$$D = L_{mod} * \cos(\beta) + [(L_{mod} * \sin(\beta)) / \tan(\alpha)]$$

A side-by-side comparison evaluated the structural performance of the 10° and 15° landscape tilt choices:

Engineering Layout Metric	Option A: 10° Landscape Tilt	Option B: 15° Landscape Tilt
Horizontal Footprint Projection (P_horiz)	1.117 meters	1.095 meters
Minimum Clear Shading Walkway (D_clear)	22.3 cm (Too Tight for Cleaning Access)	33.2 cm (Optimal Clear Shading Pass)
Total Row-to-Row Pitch Distance (D)	**1.340 meters**	**1.427 meters**
Module Structural Apex Height (h)	19.7 cm (Ultra-Low Profile)	29.4 cm (Ultra-Low Profile)
Aerodynamic Net Pressure Coefficient (C_n)	-0.60 (Low Suction Lift Effect)	-0.85 (Medium Sail Suction Effect)
Wind Uplift Suction Force Per Module	**1,576.8 Newtons (~161 kg)**	**2,233.8 Newtons (~228 kg)**
Total Plant Cumulative Upward Lift	**38.58 Metric Tons**	**54.65 Metric Tons**

Engineering Access Adjustment Note: While a 10° or 15° tilt satisfies minimum shadow spacing at less than 1.5 meters row pitch, implementing the raw mathematical clear path (22cm to 33cm) makes physical cleaning walkways impossible. Therefore, the racking layout design manually overrides this boundary to a fixed 0.50m or 0.60m walkway gap, providing safe clearance for maintenance crews while remaining entirely free of winter shadows.

5. Phase 5 — Symmetrical System Balance Architecture

By choosing our absolute winter maximum capacity limit of 20 panels per string exactly, we establish a perfectly symmetrical, cost-optimized electrical schematic. Dividing our total target of 240 panels creates exactly 12 identical series strings.

Formula 5.1 — DC-to-AC Inverter Loading Ratio:

*The gross capacity of our array fields represents 240 modules * 585 W = 140.40 kWp DC. Wiring this straight into our single 110 kW AC string inverter gives us a perfect sizing configuration:*
Inverter Loading Ratio = 140.40 kWp DC / 110.00 kW AC = 1.2766 (Optimal Performance Window)

This 1.28 loading ratio acts as a strategic performance layer. It over-dimensions the DC field to compensate for real-world environmental losses (such as Dubai's high ambient heat and desert dust soiling), flattening the generation curve so the inverter runs at its peak 110 kW capacity for a longer portion of the day without suffering from wasteful afternoon clipping.

Formula 5.2 — Symmetrical MPPT Input Mapping & Fuse Exemption:

According to international electrical standards (IEC 60364-7-712 and DEWA / SEWA network codes), string overcurrent protection fuses are strictly required only if 3 or more strings are combined in parallel per tracker circuit. Because our balanced array maps exactly 12 strings across the Huawei inverter's 10 independent MPPT channels, we distribute inputs symmetrically across the chassis terminals:

- **MPPT 1 and MPPT 2:** Host exactly 2 strings in parallel per tracking circuit.
- **MPPT 3 through MPPT 10:** Host exactly 1 single string per tracking circuit.

Because the maximum number of parallel paths anywhere on the machine is limited to 2, reverse fault currents can never cross panel safety thresholds. Therefore, the layout is 100% exempt from string fuse requirements under utility codes. Running direct string cable paths straight into the inverter's built-in quick-connect MC4 terminals completely eliminates external combiner box enclosures, saving substantial upfront equipment capital expenditure (CapEx) and removing rooftop thermal failure points.

6. Phase 6 — Overcurrent Protection & Balance of System (BOS) Sizing

Every individual electrical leg, sub-panel breaker, and conductor cross-sectional gauge was modeled and checked from scratch to ensure total safety compliance under peak full-load delivery.

Formula 6.1 — Three-Phase Continuous AC Output Current:

The maximum continuous operating current flowing out of the inverter terminals at peak noon generation is calculated using a standard three-phase power formula operating on a 400V grid network:
$I_{ac_nominal} = P_{ac} / (\sqrt{3} * V_{line\text{-}line} * \cos(\phi)) = 110{,}000\ W / (1.7321 * 400V * 1.0) = 158.78\ Amperes$

Formula 6.2 — Main AC Overcurrent Circuit Breaker Sizing (Breaker Q):

Applying the mandatory continuous safety-factor load multiplier of 125% sets our circuit breaker current headroom ceiling:
$I_{breaker} = 158.78\ A * 1.25 = 198.48\ Amperes \rightarrow$ *Sized to standard 200A 3-Pole AC MCCB (35kA Capacity)*

Formula 6.3 — AC Transmission Delivery Cable Cross-Section:

Rearranging three-phase voltage drop equations allows us to calculate the required wire gauge size directly from scratch to maintain transmission line losses strictly beneath a 2.0% utility ceiling limit over a standard 30-meter routing corridor:
$A = (\sqrt{3} * L * I * \rho) / V_{drop_volts} = (1.7321 * 30m * 158.78A * 0.0175) / 8.0V = 18.05\ mm^2$

While a thin 18.05 mm² wire clears voltage drop, we must cross-reference for current ampacity de-rating rules under extreme Dubai summer conditions. Operating inside an industrial building zone with an internal design ambient air temperature of 45°C, the thermal correction factor for XLPE cables scales capacity down (C_a = 0.866). To provide robust continuous safety headroom over the 158.78A output and prevent thermal melting, we step up to a standard commercial 4-Core 3/0 AWG (85 mm² metric equivalence) Copper Conductor (Cu/XLPE/SWA/PVC) Armored. At this size, our real-world voltage drop shrivels to an ultra-efficient 0.52%, maximizing performance.

Formula 6.4 — Control Cabinet Auxiliary AC Protection Fuses:

The data logging cubicle contains two Switch Mode Power Supplies (SMPS) providing power to the telemetry stack (15W for the SmartLogger 3000A + 7W for the Teltonika Router). Factoring in an 80% converter efficiency profile, the single-phase 230V AC loop draws a steady current of 0.12A. To absorb input transformer startup inrush surges safely without premature burning, the control loop is protected by a pair of 2A 250V AC Slow-Blow 10x38mm Cylindrical Fuses nested inside DIN-rail mounted fuse holders.

Formula 6.5 — High-Voltage Transient Overvoltage Mitigation (SPD Selection):

- **DC Surge Arrester (IEC 61643-31):** Sized to a maximum continuous operating voltage ceiling ($U_{cpv} \geq 1.1 * V_{string_max}$) to safely handle the 1,087.2V winter open-circuit peaks. The layout specifies a decentralized model deploying two (2) separate external IP65 DC Disconnect Enclosures on the roof, with each box populated with six (6) Phoenix Contact VAL-MS-T1/T2 1000DC-PV/2+V modules (Total: 12 DC SPDs), rated to a surge discharge capacity of In = 20kA / Imax = 40kA.
- **AC Surge Arrester (IEC 61643-11):** Sized to a continuous operational threshold of Uc = 385V AC in a 3+1 multi-phase clamping network to absorb backwards grid-side switching spikes safely.

7. Comprehensive Equipment Master BOQ / BOS Schedule

The following master ledger outlines the exact itemized inventory, manufacturer ratings, and physical line quantities locked into your final AutoCAD engineering drawings and project simulation deliverables:

Item	Component Category	Technical Model & Sizing Specifications	Quantities
1	Primary PV Modules	LONGi Hi-MO X6 Explorer LR5-72HTH-585W (22.64% Efficiency)	**240 Units**
2	Central String Inverter	Huawei SUN2000-110KTL-M2 (110kW AC, 10 MPPT, 20 DC Inputs)	**1 Unit**
3	DC Disconnect Enclosures	Rooftop External IP65 Polycarbonate Enclosure w/ Rotary Handle	**2 Boxes**
4	DC Overvoltage SPDs	Phoenix Contact VAL-MS-T1/T2 1000DC-PV/2+V Pluggable Modules	**12 Pieces**
5	Main AC Protection Switch	ABB SACE Tmax XT3N 200A 3-Pole 400V MCCB (Breaker Q)	**1 Unit**
6	AC Conductor Home Run	4-Core 3/0 AWG (85 mm²) Copper (Cu/XLPE/SWA/PVC) Armored	**377.0 Feet**
7	DC String Conductor	12 AWG (4 mm²) PV1-F XLPE Solar Conductor Wire Line	**2,970.8 Feet**
8	Telemetry Data Logger	Huawei SmartLogger 3000A Gateway Platform (Modbus Master)	**1 Unit**
9	Industrial Cellular Router	Teltonika RUT955 Industrial Cellular Router / Network Switch	**1 Unit**
10	Smart Billing Net-Meter	DEWA-Approved Smart Net-Meter + 3 x Current Transformers (CTs)	**1 Set**
11	Control Power Supplies	Switch Mode Power Units: 230V AC Input to 24V DC / 5V DC	**2 Units**
12	Auxiliary Control Fuses	2A 250V AC Slow-Blow 10x38mm Cylindrical Ceramic Fuses	**2 Units**
13	Racking Mounting Setup	Anodized Aluminum L1 Click-Fit Landscape Fixed-Tilt Racking	**240 Mounts**
14	Earthing Network Loop	Bare 6 AWG Copper Jumper Wire + 16 mm² Yellow/Green Core Wire	**1 Complete Loop**

8. DEWA Grid Compliance & Structural Earthing Design

A. Shams Dubai DRRG V2.0 Interconnection Layout:

The system is engineered in complete compliance with DEWA Appendix A Connection Scheme LV4 criteria, providing a cascading safety separation line: a localized AC Generator Switch at the inverter output, a motorized Interface Switch (IPS) contactor linked to protective telemetry relays, and the manual 200A General

Switch located inside the main solar distribution sub-panel. The integrated protective relay tracks undervoltage/overvoltage thresholds (ANSI 27/59 at 0.85Vn and 1.10Vn) and anti-islanding parameters (Vector Shift ANSI 78 and ROCOF ANSI 81R) to trigger an immediate automatic disconnect upon grid failure.

B. Zero-Export Active Curtailment Control Loop:

To satisfy the strict export limitation mandate, three analog Current Transformer (CT) sensing rings are positioned on the main incomer power busbars immediately downstream of the existing facility 630A MCCB incomer panel. The CT data feeds directly into the kWh Meter reading device, which translates active load currents into high-speed Modbus-RTU RS485 communication data streams. This digital stream maps directly into the COM ports of the Huawei SmartLogger 3000A. If the factory's operational load decreases below the solar production line, the SmartLogger issues an instantaneous command to the inverter, curtailing power output and maintaining complete zero-export grid integrity.

C. Equipotential Earthing System Design Basis:

To completely eliminate static buildup, structural touch-voltage hazards, and high-voltage atmospheric lightning coupling risks, an unswitched equipotential grounding path was developed. Stainless steel star-washers pierce the anodized frames of all 240 panels, bonding them directly to the aluminum racking rails. Interconnected rows are linked using bare 6 AWG copper jumper wire loops channels clamped via tinned-copper lugs. A continuous 16 mm^2 yellow/green copper core conductor runs from the inverter terminal chassis down to a dedicated solar ground ring, which connects to three 3-meter copper-clad steel ground rods driven deeply into the soil. The completed path passes safety audits with a tested grid path impedance of $R \leq 1.0$ Ohm.

9. PVSyst / HelioScope Simulation Report & Economic ROI Analysis

An 8,760-hour annual energy yield simulation was executed inside the software workspace utilizing the localized Meteonorm 8 weather matrix dataset for Al Quoz, Dubai (Coordinates: 25.1161, 55.2353).

- **Gross Modeled Capacity:** 140.40 kWp DC Field Array / 110.00 kW AC Nameplate.
- **Simulated Annual Clean Power Yield:** 252,751.5 kWh/year (252.8 MWh).
- **Elite Environmental Performance Metrics:** Specific Yield of 1,800.2 kWh/kWp/year with a system Performance Ratio (PR) of 82.3% (comfortably clearing high-temperature desert design targets).

Financial Simple Payback Period Sizing:

Because the factory's daytime operational packaging load floor (351.92 kW) easily consumes the peak solar output capability, 100% of the simulated 252,751.5 kWh annual generation is absorbed internally on-site. Applying a typical Dubai industrial commercial tariff bracket of 0.44 AED / kWh, every unit of clean energy generated directly prevents the facility from purchasing grid power, yielding concrete annual cash flow benefits:

*Annual Utility Bill Operational Savings = 252,751.5 kWh * 0.44 AED/kWh = 111,210.66 AED / Year*

Applying a standard commercial turnkey market pricing estimate of 2.20 AED per watt installed for a premium Tier-1 plant of this scale, the total upfront implementation cost computes to 140,400 Watts * 2.20 AED/W = 308,880 AED. Sizing the simple payback window yields:

Project Simple Payback Period = Total Capital Cost / Annual Savings = 308,880 AED / 111,210.66 AED = 2.77 Years

Financial Sizing Note: This manually proven 2.77-year payback period completely overrides the software's unconfigured, default global template baseline of 8.3 years. The default algorithm double-penalizes the design by utilizing oversized US labor construction index assumptions (~$2.00/W) and masking true local generation value using low un-configured baseline utility rates. Given that commercial Tier-1 solar assets possess legal performance warranties spanning 25 years, this installation represents an outstanding financial investment, delivering over two decades of free, clean energy directly to the factory floor.

10. Pre-Energization Cold Testing & Field Commissioning Checklist

Before utility field inspectors clear the system for live parallel grid connection, the installation contractor must successfully execute and log the following cold-circuit safety sequence:

1. **Grounding Continuity Validation:** Measure the exact physical impedance path from the furthest landscape module racking rail down to the main earth pit terminal rod. Total resistance must read $R \leq 1.0\ \Omega$ to confirm robust bonding and eliminate touch-potential hazards.
2. **High-Voltage DC Insulation Resistance (IR) Proofing:** Disconnect the string home runs from ports PV1–PV12. Utilizing a calibrated insulation tester, inject a 1,000V DC potential between the positive conductor-to-earth and negative conductor-to-earth lines for every individual string. Isolation values must cross a mandatory quality floor of $> 50\ M\Omega$ to prove zero wire insulation tearing or roof-membrane short circuits.
3. **Open-Circuit Polarity and String Voltage Alignment:** Measure the voltage lines of all 12 strings using a high-voltage multimeter. Verify that every single row returns a clean positive value tracking between 850V and 1,000V DC (depending on real-time solar irradiance). Any negative or mismatched readings indicate an inverse series panel connection fault or a split-string bypass problem that must be resolved before plugging into the terminals.
4. **Zero-Export Communication & Phase Test:** Verify phase rotation alignment matches perfectly across the 200A breaker Q contacts. Power up the SmartLogger and run a local ping test to verify the RS485 Modbus twisted-pair line registers a high-speed telemetry link to the smart net-meter CT loops, ensuring active curtailment safeguards are fully online.